

The Siege of Ironhold

A tale of craft, coordination and judgment

The Iron Tyrant approaches Ironhold, the last stronghold of the free lands. His siege engines shake the roads. Behind the walls, six guilds must build a force of Titan's Bastion war golems before the final assault.

The Emberborn Smiths forge the frames. The Gearwright's Consortium builds the engines. Together their components enter the Hammer and Anvil Guild, which makes each core subassembly. At the same time, the Arcane Weavers enchant the armor. The Titanwrights combine that armor with the subassembly. Finally, the Order of the Unerring Gaze inspects each Titan before it can defend the walls.

Every completed unit is one Titan. A slower guild still makes one whole unit when its capacity and supplies permit. It never makes half a Titan.

Three cycles before the assault

Early Dawn: rounds 1-5. Learn how the guilds depend on one another.

Fires of Noon: rounds 6-10. Review delays and improve coordination.

Twilight of Battle: rounds 11-15. Complete and inspect the Titans before time expires. All six guilds operate in every round. The cycle names do not change the penalty rates.

The Council's terms

Each guild starts with **10 gold**. Before round 1, the Council announces a whole-game target of approved Titans. The class may instead use a practice game with no target and compare completed output. Gold, inventory and outstanding work carry across the three cycles.

A green signal means the local process is healthy. Yellow asks others to anticipate a delay. Red stops an unsafe or failed operation so it can be addressed. A truthful warning protects the factory. A warning is never a separate fine.

If any guild's gold reaches zero or falls below it, the whole factory fails. If all guilds survive through round 15 and meet the announced target, Ironhold is defended. Only then is the richest guild named Master Guild. Ties share the title. In practice mode, report output and remaining gold without claiming a target victory.

The challenge is to protect the common mission while using scarce resources well. Will the guilds act together, or will local decisions leave Ironhold undefended?

Rules: use Game Rule Book, revised edition 3.0. This story introduces the setting; it adds no separate penalties or dice rules.

Game Rule Book

The Siege of Ironhold / Andon and Jidoka classroom game - edition 3.0

Revision status. This is a playable, reconciled classroom baseline. Conflicting source values have been resolved and missing mechanics specified. Added design choices are listed in the Review and Change Log. Game balance still needs a classroom pilot.

1. Setup and objective

Use six teams and one facilitator (the Factory Manager). Each team starts with 10 gold. Play up to 15 rounds in three cycles of five. One round represents an 8-hour day, and full capacity is two whole units. Use d6, d10 (0 means 10) and d4 dice, inventory tokens and the revised scorecards.

Before starting, record an integer target between 1 and 30 approved Titans for the whole game, or explicitly choose practice mode with no target. Thirty is a physical upper bound, not a recommended target. Select a challenging target only after a pilot. Optionally record cycle targets for discussion; they introduce no extra fines.

The factory succeeds only if every team remains above zero gold through round 15 and approved output meets the target. Bankruptcy ends play immediately after the triggering charges are posted. Finish all charges from a simultaneous batch before testing bankruptcy. No team is declared the winner of a failed factory. On success, highest remaining gold earns Master Guild; ties share it. In practice mode report output, team cash and survival.

Role	Guild	Input and output
Shop 1	Emberborn Smiths	Raw material to component 1 (frame)
Shop 2	Gearwright's Consortium	Raw material to component 2 (engine)
Shop 3	Arcane Weavers	Raw material to component 3 (enchanted armor)
Subassembly	Hammer and Anvil Guild	1 component 1 + 1 component 2 to 1 subassembly
Final assembly	Titanwrights	1 subassembly + 1 component 3 to 1 Titan
Quality inspection	Order of the Unerring Gaze	1 uninspected Titan to 1 approved Titan

Raw material supply is unlimited except for the vendor event. Start all five intermediate buffers and the approved-Titan count at zero. Gold, buffers, incident counts and repair queues never reset at cycle boundaries.

2. One round: resolve, decide, then transfer

At the start of each round, resolve due analysis tasks, post their new repair tickets, and serve up to 8 hours of each team's due repair queue. Keep unused repair hours for later rounds. A team serving repair hours uses nominal capacity 2 for its remaining time and makes no production or event roll that round.

Stage	Teams acting together	Release rule
1	Shop 1 and Shop 2	Resolve both decisions before either releases output.
2	Shop 3 and Subassembly	Subassembly uses components 1 and 2; it does not need component 3.
3	Final assembly	Uses available subassemblies and component 3.
4	Quality inspection	Approved units leave the factory and count once toward the target.

Within each stage: determine available inputs; skip rolls if the team has no complete input set or no remaining capacity. Otherwise roll production, roll an event only on partial production, choose a permitted signal, resolve the required solution, post costs, and calculate actual output. Shops need no input tokens. For two simultaneous teams, resolve events in Shop 1 then Shop 2 order, or Shop 3 then Subassembly order; output remains simultaneous.

Production d6: 1-3 gives capacity 1; 4-6 gives capacity 2. Capacity is a limit, not guaranteed output. A partial result identifies a local incident. Lack of inputs alone does not create an event or fine.

Time and output: usable capacity = $\max(0, \text{base capacity} - \text{hours served this round} / 4)$. Hours are multiples of 4 and cannot exceed 8 served per team per round. Actual output = minimum of usable capacity and each required input-buffer quantity. For shops, output equals usable capacity. Consume one of each required input per output unit. Add output to the next buffer; retain unused inputs for later rounds. Quality output is approved output, not an extra production count.

A new immediate repair is added to the due queue and served from any of that team's remaining 8-hour allowance before its output is released. If it has already acted, serve the repair from the next round instead. Sum queued hours sequentially; do not overlap them. A zero-hour fix does not restore the lost unit from a partial roll. Only transferred tokens are good units in this simplified model.

At the end of rounds 5 and 10, pause to compare delays, signals and output. Continue the same game state. After round 15, settle outstanding analysis tasks and all uncharged consequences before final scoring; there are no extra production rounds.

3. Events and honest signals

d10	Event	Applicable teams
1, 3, 5, 7, 9	Unknown cause	Facilitator rolls a second d10 until an applicable even result occurs.
2	Machine failure	All teams
4	Defective part	Subassembly, final assembly, quality
6	Machine misalignment	All teams
8	Defective raw material	Shops 1-3 only, outside vendor protection
10	Machine breakdown	All teams

Reroll an inapplicable event; do not convert it into a free pass. During vendor protection, reroll 8, including during unknown-cause resolution. Log the final cause and all dice results. Facilitators may narrate the event but cannot invent a different monetary penalty.

Signal	When allowed	Consequence
Green	Healthy local process, no new incident	No solution roll. Input shortage is separately recorded as Idle.
Yellow - future action (YF)	Machine failure or misalignment	Roll the solution now. Charge its ticket now; serve repair hours next round.
Yellow - analysis (YA)	Machine failure or misalignment	No solution now. At the next round start resolve the analysis table below.
Red	Any non-vendor incident; mandatory for breakdown or defective part	Resolve an applicable solution now and apply its repair or rework timing.
Vendor / Idle	Vendor event, or no inputs / no available time	Use the vendor rule, or record the reason for idleness. No extra signal fee.

Red is a request for immediate action, not an automatic extra 8-hour deduction. YF is a scheduled repair, not an automatic extra lost day. Record the selected solution once. Machine breakdown must be addressed, and a known defective part cannot be waved through with green or postponed.

Analysis resolution (new explicit classroom rule)

At the start of the next round, roll d6: 1-2 = transient issue, close it with 0 hours and 0 gold; 3-4 = repair with 4 hours and 1 gold; 5-6 = escalation with 8 hours and 2 gold. Charge once and serve the repair before that round's production. The original incident is then closed. Newly rolled incidents are separate. If YA is chosen in round 15, resolve it in final settlement and deduct the cost before judging success.

This table replaces unspecified future consequences. It is a design choice for repeatable facilitation, not a probability estimate derived from the original documents.

4. Solutions, ownership and accounting

d4	Solution	Applies to	Ticket
1	Stop and investigate	Failure or misalignment	4 hours / 1 gold
2	Root cause analysis	All non-vendor incidents	8 hours / 2 gold
3	Software / parts fix	Failure or misalignment	0 hours / 0 gold
4	Rework	Defective part only	8 hours / 2 gold to origin team

Reroll inapplicable solutions until one applies. This replaces the source rule that could turn an immediate stop into indefinite analysis. Failure and misalignment have three equally likely valid solutions (1, 2, 3); defective part has two (2, 4); breakdown has only solution 2. Vendor events bypass d4 entirely.

For solutions 1-3, the detecting team owns the ticket. Red serves hours immediately if the team has not released output; YF serves them next round. Solution 3 has no hours to serve and closes the incident immediately. It still leaves this round's partial base capacity at 1.

Rework (solution 4): set the detecting team's output to zero for this round and retain its input tokens. It loses this round's production opportunity but pays no additional repair fee. Determine the origin among its immediate suppliers: for Subassembly, d6 1-3 = Shop 1 and 4-6 = Shop 2; for Final Assembly, d6 1-3 = Subassembly and 4-6 = Shop 3; for Quality, the origin is Final Assembly without a roll. Charge that origin 2 gold immediately and queue 8 hours from the next round, since it has already acted. Close the detecting incident after this hold. This is a simplified attribution exercise, not physical tracing of a bad batch.

Ticket duration	Gold charged once	Use
0 hours	0	Free fix / transient analysis
4 hours	1	Investigation / analysis
8 hours	2	Root cause / rework / voluntary vendor change
32 hours	7	Forced vendor change only

Do not calculate cost from total downtime. Two 4-hour tickets cost 2 gold; four separate 8-hour tickets cost 8 gold. One forced-vendor 32-hour ticket costs 7 gold. Keep ticket counts by category. Charge when a ticket is issued, even when its hours occur later. Do not charge again as the hours are served. The 7-gold rate is the original rule book's special package rate, not a linear conversion.

Record gold as opening gold minus all ticket charges, including pending repairs. A negative ledger balance is allowed as a record of the triggering loss; it means bankruptcy, not permission to continue. Input starvation and the partial-capacity roll reduce output but are not additional cash penalties. The workbook calculates cash, not the full game state.

5. Defective raw material and vendor changes

Only a valid event result 8 can trigger defective raw material. No solution-die face produces this event. Keep one shared occurrence counter for the current vendor across all three shops. Each accepted event counts once, even if another shop has an event in the same round.

After the first or second occurrence, the three shops vote immediately. Two or more votes for a change trigger a voluntary change: issue one 8-hour / 2-gold ticket to each shop. Otherwise keep the vendor, keep the occurrence count, and the affected shop retains its partial capacity 1 this round. There is no additional local repair roll or fee for that raw-material event.

On the third accepted occurrence, force a change without a vote: issue one 32-hour / 7-gold ticket to each shop. Do not add the voluntary charge or an extra repair for the same event. Post all three charges together, then check bankruptcy. Queue vendor-change hours for all three shops from the next round; current-round production already underway is not reversed.

On any vendor change, reset the occurrence counter to zero and activate protection immediately for the rest of that round and the next five numbered rounds. If changed in round r , reject raw-material events through $r+5$; they become eligible again in $r+6$. A change in round 4 protects the remainder of round 4 and rounds 5-9; round 10 is eligible. The protection is not permanent. A new vendor starts a new counter.

Resolve simultaneous shop events in the fixed order in section 2. If the first event causes a change, reroll a later raw-material result from that stage under the newly active protection. This prevents multiple changes being charged for the same simultaneous batch.

6. Records and end-of-game settlement

Each team records: round, production roll (or no roll), event, signal, solution and owner, new 4/8/32-hour ticket counts, hours actually served, good units transferred, pending work and closing gold. The Factory Manager also keeps the five buffer balances, approved output, shared vendor counter and protection end round.

Complete each row only after its tickets and output are final. For a round ended early by bankruptcy, record only actions already resolved. Use the settlement row for round-15 analysis and any other uncharged consequence. Already charged repair hours beyond round 15 are disclosed as outstanding hours and are not charged again. They do not create extra shipments.

The supplied spreadsheet is one team's ledger; save a separate copy for each of the six teams. Its costs and balances calculate automatically, but dice validity, queues, buffers, output and overall factory victory remain facilitator-controlled. The revised PDF scorecard includes a factory control sheet.

7. Debrief

Which shortage propagated furthest? Did gold preservation conflict with factory output? When did immediate action protect downstream work? What did random blame teach, and where would real traceability be better? What changes would make this model closer to actual Jidoka?

This beginner edition models defects as incidents that reduce or hold output. It does not simulate hidden defective units, inspection accuracy, recurring unresolved defects, labor cost, revenue or demand arrivals. Use it to discuss Andon, coordination and stopping decisions; do not interpret its random outcomes as empirical lean-manufacturing evidence.